IProPBio - Integrated Process and Product Design for Sustainable Biorefineries

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Bio-based economy involves in Europe 22 million people and turns over roughly 2.4 billion €. The full realization of its huge potential, however, requires expert knowledge and synergy of different competencies. The overall goal of IProPBio is to exchange complementary theoretical and experimental knowledge of research staff while looking for innovative answers in the field.IProPBio addresses key engineering, thermodynamic, energy, environmental, safety, production and process challenges in the design, optimization and operation of sustainable biorefinery. IProPBio will significantly impact: a) the competitiveness of EU bioeconomy; b) participants' potential and new carrier perspectives; c) exchange and transfer of high-quality multidisciplinary knowledge, advanced expertise, research and innovation between academic and non-academic participants in EU member states and third countries through the dissemination of the results achieved to target groups and the general public.

Introduction

IProPBio-Integrated Process and Product Design for Sustainable Biorefineries is a project which has received funding from the European Union's Horizon 2020 research and innovation programmeunder the Marie Skłodowska-Curie grant agreement N°778168. The IProPBio project spans over 4 years (2018-2021) and 10 countries.

IProPBio addresses key engineering, thermodynamic, energy, environmental, safety, production and process challenges in the design, optimization and operation of sustainable biorefinery systems for multiproduct portfolios. Biorefinery (BioRef) is a novel industrial concept, recognized as the most promising route for employment of the full potential of the biomass industry to utilize biowastes and industrial crops. The primary objective of biorefinery systems is to maximize the conversion of biomass and waste biomasses into high value products, to reduce the environmental pollution, caused by traditional oil-based chemical industries and to stimulate regional and local development.In the EU, BioRefs social and economic benefits are well recognized. It has been estimated that this sector has the potential to create € 32.3 trillion revenues and 1 million new jobs in Europe by 2020. Hence, the topic of our Action is timely and in complete accord with the EU and MSCA RISE priorities.

Overview of the project

A complex BioRef employs multiple alternative feedstocks, unit operations and synthetic routes, using physicochemical, chemical and biological processes to obtain, separate and purify the bioproducts. In comparable terms, the process system engineering of a sustainable BioRef is more complex than the oil refinery counterpart, due to the significant differences between petroleum and biomass feedstocks. Some of the current challenges to the development of sustainable BioRef systems with multiproduct portfolios are summarized hereunder: a) Design flexible product-tailored processes for a variety of biomass feedstocks and/or their mixtures, integrating chemical and biochemical routes into sustainable biorefining of given feed stocks; b) Relate biomass extraction and separation processes with the properties of the desired products and the sustainable utilization of the depleted matrices; c) Identify, design and use inter-changeable processes, capable of being realized according to market requirements with the same industrial equipment (reactors, separators, etc.); d) Integrate production processes (e.g., extraction, pyrolysis and gasification) to develop closed loop production with no or minimum waste, and ecocompatible impact on environment; e) Ensure a constant flow of biomass feedstocks with significantly different properties for a not interrupted production plant of a competitive size, considering also the low chemical and microbiological stability of feeds, intermediates and products; f) Identify, evaluate and design a competitive portfolio of synthetic biochemicals, based on available feedstocks and given production infrastructure.

The IProPBio project faces the above formulated and other relevant challenges. With this aim,theIProPBio Consortium incorporatesentities from the academic and non-academic sector from EU countries (Denmark - coordinator, Bulgaria, Greece, Italy, Portugal, Spain, UK) and participants from Third Countries (3 entities from the academic sector). The research groups from organizations located in the Third Countries (TCs)

are not eligible for funding and include one from Mexico, one from the USAand one from Brazil. The EU no-academic partner, PANAX-The "Panax – Homeopathy, Phytotherapy and Aromatherapy Laboratory" has extensive expertise and competencies in herbal and essential oil properties, on their therapeutic effects, on the methods and the conditions for their isolation and on their commercialization and marketing. The company has over 30 years of experience on the development and enhancement of processes for production and isolation of essential oils by purely physical mass-transfer operations, like fractional distillation.

To develop an advanced process systems engineering approach for integrated process and product design (PPD)of sustainable and competitive "smart" BioRefs, with maximized profits and minimized wastes through intensive exchange of complementary theoretical knowledge and experimental expertise of the partners, six specific objectives have been identified and converted into work packages (WPs), presented in Figure 1. The specified objectives (SOs) can be summarized as:

1. Selection of alternative feed stocks to generate a variety of high added value bio-products. The selection of alternative feedstocks for the "smart" BioRef" is a key factor of its integrated PPD, which influences the processes and equipment needed for their transformation, the products obtainable and their functional properties, as well as its impact on the environment.
2. Design, validation and implementation of a thermodynamic framework (TMF). The proper design of BioRef systems requires the application of a next generation TMF that overcomes the existing challenges, with the potential to improve the accuracy and provide high predictive ability and reliable description of the thermodynamics and kinetics of the processes involved (in biomass conversion routes and separation and purification technologies) at a wide range of operating conditions.
3. Estimation of properties for integrated process and project design. This specific objective (SOs) addresses another key factor for the realization of IProPBio, as the property values of the selected feedstocks and/or their mixtures are needed in the modeling of the transformation processes (e.g., the TMF), in the formulation of product portfolios and in the evaluation of the pollutants generated.
4. Systematic synthesis of process alternatives. This objective addresses the definition of a novel methodology able to generate all feasible flowsheets related to the purification of the products identified in the objective one.
5. Multi-objective optimization problem. Process integration and intensification of multiproduct portfolio BioRefs involve several numerical challenges. For example, the mass and energy integration in BioRefs is fundamental to reduce waste streams, to minimize energy consumption and to reduce operating costs.
6. Process Integration and Life cycle analysis of different target products. Process integration involves setting targets for best theoretical performance of a process, and solves them for various alternative processes based on constraints imposed by practical considerations.

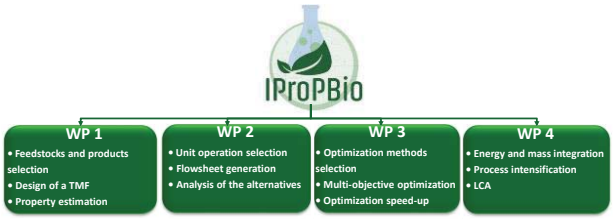


Figure 1.The project is divided four work operative packages (WPs) as reported in the diagram. To each work package is assigned a main responsible for the coordination of the work and the tasks among the participants.

In brief, the SOs1, 2 and 3 are targeted in WP1. In this work package: (1) a meticulous research of potential feedstocks will be performed. Key biomass from agricultural and other activities, which are preferably encountered at least in two different countries and the traditional/contemporary methods of biomass production and waste treatment will be charted.SO-4 is explored in WP2. The synthesis of the process alternatives regards the definition of a clear methodology used to define the research space that includes all feasible flowsheets and avoids a random alternative generation or limiting the analysis to configuration already proposed in the literature and readapted to the case examined.SO-5 is executed through WP3. In this part of the work, the appropriate design methods and optimization procedures to quantify the performances of the configurations developed in the WP2 will be defined. BioRefs show a number of challenges related to energy and water integration. Except for thermochemical conversions, they typically operate at relatively low temperatures, which complicate energy integration, involving large amount of water over several stages of the process, from pre-treatment to the synthesis and final product separation. All those problems will be addressed in the WP4.

The ultimate achievement of the successful completion of the IProPBio project include novel research results and advanced knowledge that will lead to considerable societal benefits and could potentially contribute towards practical solutions of existing societal challenges. In view of this, the IProPBio foreground knowledge will be disseminated in an optimum way for impact and re-use.

For more information:
Site of the project: http://ipropbio.sdu.dk/
EU Information:
https://cordis.europa.eu/project/rcn/212228_en.html
Research Gate:
https://www.researchgate.net/project/IProPBio-H2020-MSCA-RISE-2017-Project-number-778168
Twitter: https://twitter.com/IProPBio

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